ordering. Finally, if y is distance two or more from x, we define $x \diamond y$ to be the neighbor of x that is closest to y. One can then check that this model satisfies (5) but not (6).

Remark 5.10. These constructions are necessarily infinitary in nature, because E206 and E1648 can be shown to be equivalent for finite magmas. Indeed, E206 can be written as $x = R_y L_x L_x y$, which implies that R_y is surjective, hence injective, on a finite magma; writing $x = R_y z$ we conclude that $R_y z = R_y L_{z\diamond y} L_{z\diamond y} y$ and hence $z = L_{z\diamond y} L_{z\diamond y} y$, giving E1648. The opposite implication is similar (using E1648 to show that R_y is injective, hence surjective), and is left to the reader.

Some refutations $\mathrm{E} \not\models \mathrm{E}'$ are "immune" to translation-invariant models, in the sense that any translation-invariant model that satisfies E, also satisfies E′. One obstruction is that for such models, the squaring map S is necessarily an invertible map, since $Sx = x + f(0)$ in the abelian case and $Sx = xf(1)$ in the non-abelian case. On the other hand, adding the assumption of invertibility of squares can sometimes force the implication $\mathrm{E} \models \mathrm{E}'$ to hold. For instance, the commutative law $\mathrm{x} \diamond (\mathrm{y} \diamond \mathrm{y}) \simeq (\mathrm{y} \diamond \mathrm{y}) \diamond \mathrm{x}$ (E4482) for a square and an arbitrary element will imply the full commutative law E43 for translation-invariant models due to the surjectivity of S, but does not imply it in general (as one can easily see by considering models where S is constant).

5.4. **The twisting semigroup.** Suppose one has a magma $\mathcal{M}$ satisfying a law E, that also enjoys some endomorphisms $T, U \colon \mathcal{M} \to \mathcal{M}$. Then one can "twist" the operation $\diamond$ by T, U to obtain a new magma operation

$$x \diamond' y := Tx \diamond Uy. \tag{7}$$

If one then tests whether this new operation $\diamond'$ satisfies the same law E as the original operation $\diamond$, one will find that this will be the case provided that T, U satisfy a certain set of relations. The semigroup generated by formal generators T, U with these relations will be called the *twisting semigroup* $\mathrm{Twist}_{\mathrm{E}}$ of E. This can be best illustrated with some examples.

Example 5.11. We compute the twisting semigroup of $\mathrm{x} \simeq (\mathrm{y} \diamond \mathrm{x}) \diamond (\mathrm{x} \diamond (\mathrm{z} \diamond \mathrm{y}))$ (E1485). We test this law on the operation (7), thus we consider whether

$$x = (y \diamond' x) \diamond' (x \diamond' (z \diamond' y))$$

holds for all $x, y, z \in M$. Substituting in (7) and using the homomorphism property repeatedly, this reduces to

$$x = (T^2 y \diamond TUx) \diamond (UTx \diamond (U^2 Tz \diamond U^3 y)).$$

If we impose the conditions $TU = UT = \mathrm{id}$, $T^2 = U^3$, then this equation would follow from E1485 (with x, y, z replaced with $TUx = UTx = x$, $T^2 y$, and $U^2 Tz$ respectively). Thus the twisting semigroup $\mathrm{Twist}_{\mathrm{E1485}}$ of E1485 is generated by two generators T, U subject to the relations $\mathrm{TU} = \mathrm{UT} = 1$, $\mathrm{T}^2 = \mathrm{U}^3$. This is a cyclic group of order 5, since the relations can be rewritten as $\mathrm{T}^5 = 1$, $\mathrm{U} = \mathrm{T}^{-1}$.

Now consider $\mathrm{x} \simeq (\mathrm{x} \diamond \mathrm{x}) \diamond (\mathrm{x} \diamond \mathrm{x})$ (E151). Applying the same procedure, we arrive at

$$x = (T^2 x \diamond TUx) \diamond (UTx \diamond U^2 x)$$

so the twisting group $\mathrm{Twist}_{\mathrm{E151}}$ is generated by two generators T, U subject to the relations $\mathrm{TU} = \mathrm{UT} = \mathrm{T}^2 = \mathrm{U}^2 = 1$. This is a cyclic group of order 2, since the relations can be rewritten as $\mathrm{T}^2 = 1$, $\mathrm{U} = \mathrm{T}$.

Suppose the twisting semigroup $\mathrm{Twist}_{\mathrm{E}}$ is not a quotient of $\mathrm{Twist}_{\mathrm{E}'}$, in the sense that the relations that define $\mathrm{Twist}_{\mathrm{E}'}$ are not satisfied by the generators of $\mathrm{Twist}_{\mathrm{E}}$. Then one can often disprove the implication $\mathrm{E} \models \mathrm{E}'$ by attempting the following procedure.

- First, locate a non-trivial magma $\mathcal{M} = (M, \diamond)$ satisfying the law E. Then the Cartesian power $M^{\mathrm{Twist}_{\mathrm{E}}}$ of tuples $(x_W)_{W \in \mathrm{Twist}_{\mathrm{E}}}$, with the pointwise magma operation, will also satisfy E.
- Furthermore, this Cartesian power admits two endomorphisms T, U defined by
$$T(x_W)_{W \in \mathrm{Twist}_{\mathrm{E}}} = (x_{W\mathrm{T}})_{W \in \mathrm{Twist}_{\mathrm{E}}}; U(x_W)_{W \in \mathrm{Twist}_{\mathrm{E}}} = (x_{W\mathrm{U}})_{W \in \mathrm{Twist}_{\mathrm{E}}},$$
which satisfy the relations defining $\mathrm{Twist}_{\mathrm{E}}$.
- We now twist the magma operation $\diamond$ on $M^{\mathrm{Twist}_{\mathrm{E}}}$ by T, U to obtain a new magma operation $\diamond'$ defined by (7), that will still satisfy law E.
- Because T, U will not satisfy the relations defining $\mathrm{Twist}_{\mathrm{E}'}$, it is highly likely that this twisted operation will not satisfy E′, thus refuting the implication $\mathrm{E} \models \mathrm{E}'$. If M and the twisting semigroup were finite, this approach should also refute $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}'$.

For instance, a non-trivial finite model for E1485 is given by the finite field $\mathbb{F}_2$ of two elements with the NAND operation $x \diamond y \coloneqq 1 - xy$. If we twist $\mathbb{F}_2^5$ by the left shift $T(x_i)_{i=1}^5 = (x_{i+1})_{i=1}^5$ and right shift $U(x_i)_{i=1}^5 = (x_{i-1})_{i=1}^5$, where we extend the indices periodically modulo 5, then the resulting operation
$$(x_i)_{i=1}^5 \diamond' (y_i)_{i=1}^5 \coloneqq (1 - x_{i+1} y_{i-1})_{i=1}^5$$
on $\mathbb{F}_2^5$ still satisfies E1485, but does not satisfy E151, thus showing that E1485 $\not\models_{\mathrm{fin}}$ E151 and hence E1485 $\not\models$ E151. This particular implication does not seem to be easily refuted by any of the other methods discussed in this paper.

5.5. **Greedy constructions.** We have found *greedy extension methods*, or *greedy methods* for short, to be a powerful way to refute implications, especially when the carrier M is allowed to be infinite. Such constructions have a long history in model theory, with possibly the earliest[9] such construction due to Skolem [63]. A basic implementation of this method is as follows. To build a magma operation $\diamond \colon M \times M \to M$ that satisfies one law E but not another E′, one can first consider *partial magma operations* $\diamond \colon \Omega \to M$, defined on some subset Ω of $M \times M$. Thus $x \diamond y$ is defined if and only if $(x, y) \in \Omega$. A magma operation is then simply a partial operation which is *total* in the sense that $\Omega = M \times M$. We say that a partial magma operation is *finitely supported* if Ω is finite.

In the language of first-order logic (in which functions and relations must be total), it is convenient to view a partial magma operation as a ternary relation $R(x, y, z)$ on M with the axiom that $R(x, y, z) \wedge R(x, y, z') \implies z = z'$ for all $x, y, z \in M$. The support Ω is then the set of (x, y) for which $R(x, y, z)$ holds for some (necessarily unique) z, which one can then take to be the definition of $z = x \diamond y$.

We say that one partial operation $\diamond' \colon \Omega' \to M$ *extends* another $\diamond \colon \Omega \to M$ if Ω' contains Ω, and $x \diamond y = x \diamond' y$ whenever $x \diamond y$ (and hence $x \diamond' y$) are defined. Given a sequence $\diamond_n \colon \Omega_n \to M$ of partial operations, each of which is an extension of the previous, we can define the *direct limit* $\diamond_\infty \colon \bigcup_n \Omega_n \to M$ to be the partial operation defined by $x \diamond_\infty y \coloneqq x \diamond_n y$ whenever $(x, y) \in \Omega_n$.

[9] We thank Stanley Burris for this reference.

Abstractly, the greedy algorithm strategy can now be described as follows.

Theorem 5.12 (Abstract greedy algorithm)**.** *Let* E, E' *be equational laws, and let* Γ *be a theory of first-order sentences regarding a partial magma operation* $\diamond\colon \Omega \to M$ *on a carrier* M*. Assume the following axioms:*

(i) (Seed) There exists a finitely supported partial magma operation $\diamond_0\colon \Omega_0 \to M$ *satisfying* Γ *that contradicts* E'*, in the sense that there is some assignment of variables in* E' *in* M *such that both sides of* E' *are defined using* $\diamond_0$*, but not equal to each other.*

(ii) (Soundness) If $\diamond_n\colon \Omega_n \to M$ *is a sequence of partial magma operations satisfying* Γ *with each* $\diamond_{n+1}$ *an extension of* $\diamond_n$*, and the direct limit* $\diamond_\infty$ *is total, then this limit satisfies* E*.*

(iii) (Greedy extension) If $\diamond\colon \Omega \to M$ *is a finitely supported partial magma operation satisfying* Γ*, and* $a, b \in M$*, then there exists a finitely supported extension* $\diamond'\colon \Omega' \to M'$ *of* $\diamond$ *to a possibly larger carrier* M'*, and also satisfying* Γ*, such that* $a \diamond' b$ *is defined.*

Then $\mathrm{E} \not\models \mathrm{E}'$*.*

We remark that this greedy method seems to be inherently infinitary in nature, and does not seem well adapted to refute finite magma implications $\mathrm{E} \models_{\mathrm{fin}} \mathrm{E}'$.

Proof. We work on the countably infinite carrier $\mathbb{N}$. By embedding the finitely supported operation $\diamond_0$ from axiom (i) into $\mathbb{N}$, we can assume without loss of generality that $\diamond_0$ has carrier $\mathbb{N}$. By similar relabeling, we can assume in (iii) that $M' = M$ when $M = \mathbb{N}$, since any elements of $M' \setminus \mathbb{N}$ that appear in Ω' can simply be reassigned to natural numbers that did not previously appear in Ω. We well-order the pairs in $\mathbb{N} \times \mathbb{N}$ by (a_n, b_n) for $n = 1, 2, \dots$. Iterating (iii) starting from $\diamond_0$, we can thus create a sequence of finitely supported magma operations $\diamond_0, \diamond_1, \dots$ on $\mathbb{N}$ satisfying Γ, with each $\diamond_{n+1}$ an extension of $\diamond_n$, and $a_n \diamond_n b_n$ defined for all $n \geq 1$. Then the direct limit $\diamond_\infty$ of these operations is total, and does not satisfy E' thanks to axiom (i). On the other hand, by axiom (ii) it satisfies E, and the claim follows. □

We refer to Γ as the *rule set* for the greedy extension method. To apply Theorem 5.12 to obtain a refutation $\mathrm{E} \not\models \mathrm{E}'$, we have found the following trial-and-error method to work well in practice:

1. Start with a minimal rule set Γ that has just enough axioms to imply the soundness property for the given hypothesis E.
2. Attempt to establish the greedy extension property for this rule set by setting $a \diamond' b$ equal to a new element $c \notin M$, and then defining additional values of $\diamond'$ as necessary to recover the axioms of Γ'.
3. If this can be done in all cases, then locate a seed $\diamond_0$ refuting the given target E', and `STOP`.
4. If there is an obstruction (often due to a "collision" in which a given operation $x \diamond' y$ is required to equal two different values), add one or more rules to Γ to avoid this obstruction, and return to Step 2.